

Pradeep Somasundaram

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SOFTWARE ENGINEER — AI, SEARCH & DISTRIBUTED SYSTEMS

Software Engineer / AI Engineer experienced in building production-grade backend, distributed, and ML systems using Python, Linux/Unix, REST/gRPC, TensorFlow, Spark, GCP, AWS, Docker, and Kubernetes. Built high-volume distributed pipelines processing **10M+ records monthly**, near-real-time ingestion services, production APIs, and cloud-native applications. Experienced in information retrieval, semantic/vector search, RAG, embeddings, NLP, deep learning, and distributed systems. Strong background in production troubleshooting, observability, performance optimization, and scalable data processing. M.S. in Data Science with graduate training in Machine Learning, NLP, Deep Learning, Statistical Learning, Optimization, and Time Series.

TECHNICAL SKILLS

Programming & Systems: Python, C / C++, SQL, Bash, Object-Oriented Programming, Data Structures & Algorithms, Linux/Unix, Concurrent Processing, Git

Distributed & Real-Time Systems: Apache Spark, PySpark, GCP Dataflow, Kafka, Batch Processing, Streaming Pipelines, Event-Driven Processing, Distributed Computing, ETL/ELT, Fault-Tolerant Pipelines, Low-Latency Data Delivery

Search, Retrieval & AI: Information Retrieval, Semantic Search, Vector Search, Embeddings, Retrieval-Augmented Generation (RAG), Query Understanding, Search Relevance, Retrieval Evaluation, Pinecone, LangChain, LangGraph, Large Language Models

Machine Learning: TensorFlow, Deep Learning, Natural Language Processing, LSTM, Time-Series Modeling, Anomaly Detection, Predictive Modeling, Feature Engineering, Model Training, Model Evaluation, Validation, Inference Workflows

Backend & APIs: RESTful Services, gRPC, Microservices, API Integration, JSON, Backend Services, Service Integration, Production Support, Debugging, Performance Optimization

Cloud & Production Engineering: GCP BigQuery, GCP Dataflow, AWS S3, AWS Glue, AWS Lambda, IAM, Databricks, Docker, Kubernetes, CI/CD, GitHub Actions, Azure DevOps, Monitoring, Logging, Observability, Data Quality

PROFESSIONAL EXPERIENCE

Cognizant Technology Solutions, Dallas, TX *Data / AI Solutions Engineer*

Feb 2026 – Present

- Engineer scalable Python and SQL backend data services for a U.S. restaurant-technology modernization program, migrating legacy SQL Server/PL-SQL workloads to GCP BigQuery and Dataflow while processing **10M+ records monthly**.
- Design high-volume batch and near-real-time ingestion pipelines for API and file-based sources using distributed processing architectures, reducing downstream data latency by approximately **30%** while improving reliability and operational visibility.
- Build reusable distributed transformation frameworks across Bronze, Silver, and Gold layers with schema enforcement, automated validation, data-quality checks, fault handling, and production monitoring for high-volume workloads.
- Develop Python-based backend integrations and RESTful service workflows supporting ingestion, transformation, and low-latency delivery of production data to downstream analytical and intelligent applications.
- Optimize Python and SQL workloads, analyze runtime behavior and logs, troubleshoot production failures, and collaborate with software engineering and DevOps teams on CI/CD, deployment, monitoring, observability, and incident resolution.

Beauty Manufacturing Solutions Corp (BMSC), Dallas, TX *Agentic AI Engineer / Data Engineer*

Oct 2025 – Feb 2026

- Built distributed PySpark and Spark SQL pipelines on AWS and Databricks to migrate approximately **500 GB** of SQL Server data and process **2M+ records monthly** across manufacturing, quality, ERP, and operational systems.
- Developed scalable cloud-native ingestion and processing services using Python, AWS S3, Glue, Lambda, IAM, Databricks, and REST APIs with automated validation, retry handling, error recovery, and production data-quality controls.
- Containerized Python data and AI services using Docker and supported GitHub-based CI/CD pipelines for automated testing, repeatable deployments, debugging, and production support.
- Implemented retrieval-augmented AI workflows connecting enterprise data with LLM-based applications using embeddings, retrieval orchestration, and grounded knowledge access to improve reliability of generated responses.

Chicago Education Advocacy Cooperative, Chicago, IL (Remote) Data Scientist*Jun 2025 – Sep 2025*

- Built Python and SQL data-validation, transformation, and analytical workflows supporting programs serving **5,000+ stakeholders**, improving data quality and analytical reliability by **20%**.
- Automated repeatable data-processing and reporting workflows, reducing manual reporting effort by **25%**, while partnering with stakeholders to define evaluation metrics, validate outputs, and measure solution quality.

Shree Kay Vee Automation, Chennai, India Data Scientist*May 2023 – Jul 2023*

- Developed Python and SQL ingestion and analytical pipelines for telemetry from **100+ industrial assets**, reducing data-processing latency by **40%** and improving availability of operational signals for downstream applications.
- Built and evaluated machine-learning workflows for anomaly detection and predictive maintenance using time-series and LSTM-based models, including feature engineering, model validation, performance evaluation, and inference-oriented data preparation.
- Integrated industrial telemetry and enterprise data through APIs and cloud services while supporting debugging, testing, deployment, performance optimization, and runtime issue resolution in Linux/cloud environments.

KaaShiv InfoTech, Chennai, India AI & Data Engineer*Jan 2023 – Mar 2023*

- Built retrieval-augmented AI pipelines in Python using embeddings, Pinecone vector search, LLMs, and orchestration frameworks to retrieve contextually relevant enterprise knowledge for grounded question-answering applications.
- Developed semantic retrieval workflows combining vector representations, similarity-based retrieval, contextual query processing, and downstream LLM generation to improve relevance of retrieved enterprise information.
- Developed REST and gRPC-based service integrations connecting AI components, data stores, cloud services, and downstream applications using modular backend interfaces and reliable data-exchange patterns.
- Implemented cloud-based processing workflows using AWS Lambda and analytical data platforms while supporting testing, debugging, performance tuning, deployment, monitoring, and production troubleshooting.

IBM, Chennai, India Machine Learning Engineer*Aug 2022 – Dec 2022*

- Developed and evaluated Python-based machine-learning workflows for structured and unstructured datasets, including preprocessing, feature engineering, model training, validation, and performance analysis.
- Supported API-driven and data-processing components consumed by analytical applications and performed debugging, testing, optimization, integration, and defect resolution throughout the software development lifecycle.

Shree Kay Vee Automation, Chennai, India Data Scientist*Mar 2020 – Jun 2022*

- Designed Python and SQL data-processing and machine-learning workflows for industrial IoT use cases including time-series forecasting, predictive maintenance, anomaly detection, and operational analytics.
- Built reusable ingestion and transformation pipelines for high-volume sensor and enterprise data, integrating APIs and relational data sources while improving reliability of downstream analytics and ML workflows.
- Developed production-oriented analytical workflows and collaborated with engineering teams on requirements, implementation, testing, debugging, optimization, deployment, monitoring, and production support.

SEARCH, ML & SYSTEMS ENGINEERING

- **Information Retrieval:** Built semantic retrieval and RAG workflows using Python, embeddings, Pinecone vector search, contextual query processing, and grounded generation for enterprise knowledge applications.
- **Distributed Data Systems:** Designed scalable batch and near-real-time ingestion pipelines using Spark, PySpark, Dataflow, cloud storage, APIs, distributed processing patterns, fault handling, and production monitoring.
- **Machine Learning Systems:** Developed model-training and evaluation workflows spanning NLP, deep learning, LSTM-based time-series modeling, anomaly detection, feature engineering, validation, and inference preparation.
- **Production Software Engineering:** Developed and supported REST/gRPC integrations, Python backend services, Linux/cloud applications, Dockerized workloads, CI/CD pipelines, monitoring, logging, performance optimization, and production incident resolution.

EDUCATION

The George Washington University, Washington, DC*Aug 2023 – May 2025**Master of Science in Data Science*

Graduate coursework and research in Machine Learning, Statistical Learning, Natural Language Processing, Deep Learning, Optimization, Time Series, Data Visualization, and Quantitative Analysis.

CERTIFICATION

AWS Certified Cloud Architect — Cloud architecture, security, IAM, storage, networking, serverless services, and scalable cloud solutions.